

Power Pivot Data Model Documentation

Engagement & Child Activity Analytics

Date: 9/19/2026

1. PROJECT OVERVIEW

This Power Pivot data model analyzes child engagement in activities across multiple organizations. It combines enrollment data, activity metrics, child demographics, lifestyle information, and preference data.

Model Components:

- 8 data tables (engagement, child, activity, org, preferences, child_preferences, child_org, life_style)
- 1 date table (DimDate)
- 9 active relationships + 1 inactive
- 15+ calculated columns
- 30+ DAX measures across 7 sections
- 5 KPIs with traffic-light status
- 11 pivot tables (P1-P11)

2. DATA ISSUES & WARNINGS

⚠ THREE CRITICAL ISSUES:

Issue #1: Broken Foreign Keys

Only ~6% of engagement[child_id] exist in child table. Only ~67% of engagement[activity_id] exist in activity table. Causes most sessions to show under (blank) when filtering by child attributes.

Issue #2: Mixed Units in Lifestyle Data

screen_time, gaming, social_media_usage: some rows contain minutes instead of hours. Only 776/1000 rows are clean.

Issue #3: Variant Interaction Types

activity[interaction_type] has variants: grp/Group, ind/Individual, pr/Pair/2-person.

3. RELATIONSHIPS

#	From (Many)	To (One)	Active	Purpose
1	engagement[child_id]	child[child_id]	✓	Main child link
2	engagement[activity_id]	activity[activity_id]	✓	Main activity link
3	engagement[session_date]	DimDate[Date]	✓	Time analysis
4	activity[org_id]	org[org_id]	✓	Session-to-org
5	child[child_id]	life_style[child_id]	✓	Lifestyle per child
6	child[child_id]	child_preferences[child_id]	✓	Preferences
7	child_preferences[pref_id]	preferences[pref_id]	✓	Preference map
8	child_org[child_id]	child[child_id]	✓	Enrollments
9	child_org[org_id]	org[org_id]	X	Avoid ambiguity

Note: Relationship #9 (child_org→org) is INACTIVE to avoid ambiguity. Measures using it call USERRELATIONSHIP().

4. CALCULATED COLUMNS

TABLE: child

Age Band: Groups ages into 00-06, 07-09, 10-12, 13-15, 16-18, Unknown

Income Band: Groups income into 1.Under20K, 2.20K-40K, 3.40K-60K, 4.60K+

Sleep Hours: RELATED(life_style[sleep_duration])

Outdoor Hours: RELATED(life_style[outdoor_time])
 Screen Hours (Clean): Filters screen_time by unit_mismatch_flag & tech_consistency_flag = "OK"
 Sleep Band: Groups sleep into 1.Under8h, 2.8-10h, 3.10h+
 Screen Band: Groups clean screen into 1.Under2h, 2.2-4h, 3.4h+
 Preference Name: RELATED(preferences[name]) through child_preferences
 Preference Category: RELATED(preferences[category]) through child_preferences
 TABLE: activity
 Interaction Type Clean: Standardizes grp→Group, ind→Individual, pr→Pair
 Duration Band: Groups into 1.Short(<30), 2.Medium(30-59), 3.Long(60+)
 TABLE: engagement
 Engagement Level: Levels into 1.Low(<50), 2.Medium(50-74), 3.High(75+)
 Valid Child Link: 1 if child_id exists, 0 if blank (data quality flag)
 Valid Activity Link: 1 if activity_id exists, 0 if blank (data quality flag)
 TABLE: DimDate
 Cal Year: =YEAR(DimDate[Date])
 Year-Month: =FORMAT(DimDate[Date], "YYYY-MM")
 Year-Quarter: =YEAR(DimDate[Date]) & "-Q" & ROUNDUP(MONTH(...)/3, 0)

5. MEASURES (30+ DAX FORMULAS)

Section 4.1: CORE MEASURES (12)

Total Sessions, Children Engaged, Sessions per Child, Avg Engagement Score, Avg Task Completion %, Avg Response Time, Avg Attention Shifts, Avg Attention Duration, High/Low Engagement Sessions, % High/Low Engagement

Section 4.2: BENCHMARK & COMPARISON (4)

Overall Avg Engagement (unfiltered baseline), Engagement vs Overall (variance), Avg Engagement (min 10 sessions), Org Rank (RANKX dense)

Section 4.3: TIME INTELLIGENCE (9)

Last Data Date, Avg Engagement PM/PY, MoM/YoY Changes, Sessions YTD/Last90Days, Rolling 3M Average

Sections 4.4-4.7: CHILD/EQUITY, ACTIVITY/ORG, LIFESTYLE, DATA QUALITY (20+)

Includes: Tech access metrics, Digital Divide Gap, Activity/Org counts, Lifestyle averages, Data quality KPIs

6. KEY PERFORMANCE INDICATORS (5 KPIs)

KPI	Base Measure	Target	Red	Green
1	Avg Engagement Score	70	< 56	≥ 70
2	Avg Task Completion %	65%	< 52%	≥ 65%
3	% High Engagement	20%	< 12%	≥ 20%
4	Avg Attention Shifts	10	> 12.5	≤ 10
5	% Sessions Linked to Child	95%	< 76%	≥ 95%

Status icons: Red = Below target, Yellow = At risk, Green = On target. Targets are starting suggestions; adjust to your strategy.

7. PIVOT TABLES (11 Pivots)

P1 KPI: Summary dashboard
 P2 Trend: Monthly engagement over time with chart
 P3 Activity Type: Which activities engage best
 P4 Difficulty×Format: Heatmap of format performance by difficulty
 P5 Group/Pair/Ind: Social format impact
 P6 Session Length: Duration vs attention
 P7 Age×School: Demographic heatmap
 P8 Equity&Tech: Digital divide analysis

P9 Preference×Activity: Preference matching patterns

P10 Top/Bottom Orgs: Best and worst performers (Top 10 + Bottom 10)

P11 Org Type: Performance by organization type

8. RECONCILIATION CHECKS

✓ Total Sessions (no slicers) = 1,000

✓ Total Sessions summed across any pivot = 1,000

✓ Overall Avg Engagement constant when clicking slicers

✓ Enrollments summed over all orgs = 1,000

✓ % Sessions Linked to Child $\approx$ 6% (will jump to 95%+ when IDs fixed)

9. NEXT STEPS

For Excel Dashboard:

1. Add Slicers: org[type], child[school type], child[Age Band], activity[activity type], DimDate[Cal Year]
2. Add Timeline: DimDate[Date]
3. Connect all slicers to all pivots
4. Arrange on Dashboard sheet

For Power BI Migration:

1. Open Power BI Desktop
2. File > Open > Select power_pivot.xlsx
3. All tables, relationships, calculated columns import automatically ✓
4. Recreate ~30 measures in Power BI (copy/paste DAX formulas)
5. Create visualizations (better than pivot tables)

This documentation is complete for Phase 5 (Power Pivot). Update whenever calculated columns, measures, or KPI targets change.

Version 1.0 | 9/19/2026